

NITHIN K

Erode, Tamil Nadu | Phone: 9790578877

Email: nithinjr76@gmail.com | LinkedIn: linkedin.com/in/nithin-k-994979291 | GitHub: github.com/Nithin-dot-k

Professional Summary

Software Engineer combining offensive cybersecurity knowledge with robust full-stack development skills. Experienced in engineering secure web architectures using **React.js** and **FastAPI**, alongside hands-on project experience building secure, low-level process managers. Passionate about mitigating vulnerabilities and integrating proactive security boundaries into modern applications and AI workflows.

Technical Skills

Programming & Scripting: Java, Python, Bash, JavaScript, Rust

Frontend Development: React.js, Next.js

Security & Tools: Wireshark, Ethical Hacking Methodologies, Nmap, Burp Suite

Networking: Networking Fundamentals (TCP/IP, DNS), Protocol Analysis (HTTP, FTP, Telnet, SMTP)

Security Domains: Network Security, Web Application Security, Cryptography, AI Security

Platforms: TryHackMe (Top 25%) — tryhackme.com/p/nithinjr76

Projects

ORE - Open Runtime Environment For LLMs (Ongoing)

Rust, Hugging Face Candle, Axum

- Co-developing a kernel-level process manager in Rust to securely virtualize and orchestrate local AI inference.
- Engineering a "Context Firewall" module to automatically block heuristic prompt injections and redact PII via regex.
- Implementing a semaphore-based GPU scheduler and SSD Pager system to multiplex model weights securely.
- Designing a Semantic Bus with zero-copy embedding caching for isolated Inter-Process Communication (IPC).

InsightStream - Enterprise Multi-Tenant AI Knowledge Base

Next.js, Supabase, Clerk, Groq

- Engineered a secure B2B SaaS Document Intelligence platform using Retrieval-Augmented Generation (RAG).
- Implemented cryptographic tenant isolation via Clerk Organizations and JWTs, securing PostgreSQL partitions.
- Hardened SPA security with a custom session-change listener to instantly flush in-memory states upon logout.
- Optimized vector ingestion using Cohere's API, bypassing edge-function execution timeouts for 1024-d vectors.

PhishGuard - AI-Powered Phishing Detection System

Python, FastAPI, Scikit-learn, JavaScript

- Developed a hybrid cybersecurity tool using Random Forest ML and Heuristic Analysis via a Chrome Extension.
- Engineered a multi-layer architecture including a Whitelist and Brand-impersonation engine to reduce false positives.
- Optimized system performance using FastAPI's asynchronous capabilities, achieving sub-200ms analysis latency.

Internships & Practical Experience

Intern - Cybersecurity & Ethical Hacking

June 2025 – July 2025

Hackup Technology Pvt. Ltd.

Coimbatore

- Gained hands-on experience in ethical hacking, performing vulnerability analysis using **Nmap**.
- Conducted traffic analysis of insecure communication protocols using **Wireshark** to identify data leakages.
- Deployed local web environments using **XAMPP**, gaining practical knowledge of web server security.
- Utilized **Ngrok** to securely expose local development servers to the internet using secure tunneling.

Certifications & CTF Competitions

Privacy and Security in Online Social Media — *NPTEL* — Grade 'A' (Elite + Silver)

Certification on CISSP (Certified Information Systems Security Professional) — *Red Team Leaders*

KubSTU CTF 2026 (International) — Ranked **287/1280** teams across 51 countries — May 2026

THCON 2026 & Maltego Community OSINT CTF — Active participant in global challenges

Education

Dr. N.G.P. Institute of Technology

2023 – 2027

Bachelor of Engineering in CSE (Cyber Security) | **CGPA: 7.7**

SRC Puliampatti, Erode

2021 – 2023

Secondary School Education